Reading: Course 1 resources and citations

Module 1: Welcome to the exciting world of cybersecurity

Resources

Common cybersecurity terminology

- National Institute of Standards and Technology glossary

Transferable and technical cybersecurity skills

- CompTIA Security+ exam

Citations

Introduction to cybersecurity

- Cybersecurity and Infrastructure Security Agency. (2009, May 6). What is cybersecurity?
- Dorton, D. (2022, October 12). 6 benefits of cyber security–Defending against cyber attacks. Dean Dorton - CPAs and advisors.

Common cybersecurity terminology

- National Institute of Standards and Technology. (n.d.). Glossary. Accessed December 2022.

Transferable and technical cybersecurity skills

- CompTIA. (n.d.). CompTIA Security+ exam.

Module 2: The evolution of cybersecurity

Resources

Determine the type of attack

- artificial intelligence and machine learning
- National Institute of Standards and Technology (NIST) glossary

Understand attackers

- Hacking Google

Citations

Past cybersecurity attacks

- Carnegie Mellon University. (1988). Fostering growth in professional cyber incident management. Software Engineering Institute. CERT® is now a registered service mark of Carnegie Mellon University in the field of security of computer-stored information.

Attacks in the digital age

- Cable News Network. (2020, May 3). " I love you": How a badly-coded computer virus caused billions in damage and exposed vulnerabilities which remain 20 years on . Business.
- Federal Trade Commission. (2022, September). Equifax data breach settlement . Recent FTC cases resulting in refunds.
- National Institute of Standards and Technology. (n.d.). Glossary . Accessed December 2022.

Introduction to the eight CISSP security domains, Part 1

- The International Information Systems Security Certification Consortium. (2021, May 1). CISSP certification exam outline .

Introduction to the eight CISSP security domains, Part 2

- The International Information Systems Security Certification Consortium. (2021, May 1). CISSP certification exam outline .

Determine the type of attack

- Common Attack Pattern Enumeration and Classification. (n.d.). CAPEC-55: Rainbow table password cracking . CAPEC list.
- Cybersecurity & Infrastructure Security Agency. (2011, April 28). Security tips (ST08-001): Using caution with USB drives . Cybersecurity & infrastructure security agency.
- Commonwealth of Australia 2022. ACSC. (n.d.). Dictionary attack . Glossary.
- Cybersecurity and Infrastructure Security Agency. (2019, August 8). ACSC releases advisory on password spraying attacks .
- Cybersecurity & Infrastructure Security Agency. (2019, October). E-skimming: Skimming online customer payment data from website checkout forms . National cybersecurity awareness month.
- National Institute of Standards and Technology. (2019, July). Cybersecurity for small businesses . Cybersecurity topics. Accessed December 2022.
- National Institute of Standards and Technology. (n.d.). Glossary . Accessed December 2022.
- National Institute of Standards and Technology. (n.d.). Artificial intelligence: Adversarial machine learning . National cybersecurity center of excellence. Accessed December 2022.

Understand attackers

- Google. (2022, October 7). Hacking Google .

Module 3: Protect against threats, risks, and vulnerabilities

Resources

Controls, frameworks, and compliance

- NIST CSF
- NIST RMF
- critical infrastructure
- Executive Order on Improving the Nation's Cybersecurity

Ethical concepts that guide cybersecurity decisions

- Tallinn Manual online

Citations

Introduction to security frameworks and controls

- National Institute of Standards and Technology. (n.d.). An introduction to the components of the framework . Cybersecurity framework. Accessed December 2022.

Secure design

- National Institute of Standards and Technology. (2020, December). Data integrity: Identifying and protecting assets against ransomware and other destructive events . NIST special publication 1800-25A. Accessed December 2022.
- The Open Web Application Security Project. (2021). A04:2021 - Insecure design . OWASP top ten. Accessed December 2022.

Controls, frameworks, and compliance

- Bureau of the fiscal service. (n.d.). Rules for using CAS . CAS Rules and security requirements.
- Centers for Disease Control and Prevention. (n.d.). Health insurance portability and accountability act of 1996 (HIPAA). Public health law.
- Department of Health and Human Services. (2020, September 3).CIS controls and the HPH . HHS cybersecurity program.
- Federal Aviation Administration. (n.d.). International standards organization . Mike Monroney aeronautical center.
- FedRamp. (n.d.). Securing cloud services for the federal government . Information for our partners.
- Gov.uk. (n.d.). Data protection . Your rights and legal support.
- National Institute of Standards and Technology. (n.d.). Glossary . Accessed December 2022.
- National Institute of Standards and Technology. (2018). An introduction to the components of the framework . Cybersecurity framework. Accessed December 2022.
- National Institute of Standards and Technology. (n.d.) Beginning our journey to the NIST cybersecurity framework (CSF) . Cybersecurity framework. Accessed December 2022.
- National Institute of Standards and Technology. (2020, December). Data integrity: Identifying and protecting assets against ransomware and other destructive events . NIST special publication 1800-25A. Accessed December 2022.
- NDNB. (n.d.). SOC 1 (SSAE 16/SSAE 18) - New standards for SOC 1 reports . Compliance white papers.
- The Open Web Application Security Project. (2021). A04 insecure design . OWASP top ten. Accessed December 2022.
- The White House. (2021, May 12). Executive order on improving the nation's cybersecurity . Presidential actions.
- United States Environmental Protection Agency. (n.d.). U.S. grid regions . U.S. electricity grid & markets.

Ethics in cybersecurity

- Vallor, S., Rewak, W.J. (n.d.). An introduction to cybersecurity ethics . Santa Clara university.

Ethical concepts that guide cybersecurity decisions

- 113th Congress. (2014, July 10). Summary: S.2588—113th congress . Cybersecurity information sharing act of 2014.
- Bureau of Justice Assistance. (n.d.). Electronic communications privacy act of 1986 . Justice information sharing.
- Centers for Disease Control and Prevention. (2022, June 28). Health insurance portability and accountability act of 1996 . Public Health Professionals Gateway.
- Congressional Research Service. (2020, September 21). Cybercrime and the law: Computer fraud and abuse act (CFAA) and the 116th congress . Congressional research service.

Module 4: Cybersecurity tools and programming languages

Resources

Tools for protecting business operations

- Threat Horizon Report
- Free Cybersecurity Services and Tools

Use tools to protect business operations

- Open Web Application Security Project (OWASP) Top 10

Create a cybersecurity portfolio

- Get started with GitHub

Citations

Common cybersecurity tools

- Google Cloud. (n.d.). Chronicle overview. Guides.
- National Institutes of Health. (2021, July 21). Security information and event management (SIEM): Analysis, trends, and usage in critical infrastructures. Journal list.
- Splunk. (n.d.). Why splunk? Products.

Tools for protecting business operations

- Cybersecurity & Infrastructure Security Agency. (n.d.). Free cybersecurity services and tools. Cybersecurity.
- Cybersecurity Action Team. (2022, September). Threat Horizons. Cybersecurity action team.
- International Association of Chiefs of Police. (n.d.). Understanding digital evidence. Law Enforcement Cyber Center.
- National Institutes of Health. (2021, July 21). Security information and event management (SIEM): Analysis, trends, and usage in critical infrastructures. Journal list.
- National Institutes of Health. (2022, February). Chain of custody. PMC PubMed Central.

Use tools to protect business operations

- Auth0. (2022, March, 2). Encoding, encryption, and hashing. Blog.
- National Institute of Standards and Technology. (n.d.). Glossary. Accessed December 2022.
- The Open Web Application Security Project. (2022). OWASP top ten. Main. Accessed December 2022.

Create a cybersecurity portfolio

- GitHub. (n.d.). About READMEs. Customize your repository.
- GitHub. (n.d.). Basic writing and formatting syntax. Start writing on GitHub.
- Markdown Guide. (n.d.). The markdown elements outlined in the original design document. Basic syntax.
- GitHub. (2022, March 27). Markdown cheatsheet. Wiki.
- GitHub. (n.d.). Signing up for a new GitHub account. Get started.
- Markdown Guide. (n.d.). A quick reference to the markdown syntax. Markdown cheat sheet.
